

Represent the given C++ code accordingly: (5+6+5+7)

Test duration: 40 minutes.

- 1 **Class Diagram:** Show classes, attributes, methods, and relationships.
- 2 **Sequence Diagram:** Depict a scenario where a Member borrows books and views the borrowed list. Use named objects.
- 3 **Use Case Diagram:** Describe actions and operations seen in the code.
- 4 **Use Case Textual representation:** Describe actions from the perspective of the Member role (based on the operations seen in the code).

[Click for Example C++ code](#)

Reasoning your choice, construct Use Case diagram. (2)

For an **Online Shopping System**, following use cases are defined:

- Place Order, Auth User, Make Payment, Apply Discount, Track Order.

Which of the following represents most semantically correct use of `<<include>>` (or `<<in>>`) and `<<extend>>` (or `<<ex>>`) relationships?

- 1 Place Order `<<in>>` Auth User, Place Order `<<ex>>` Apply Discount, Track Order `<<in>>` Make Payment
- 2 Auth User `<<ex>>` Place Order, Make Payment `<<in>>` Track Order, Place Order `<<ex>>` Apply Discount
- 3 Place Order `<<in>>` Make Payment, Place Order `<<in>>` Auth User, Place Order `<<ex>>` Apply Discount
- 4 Track Order `<<ex>>` Make Payment, Apply Discount `<<in>>` Place Order, Auth User `<<in>>` Make Payment